

Jayjeet Sarkar

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PROFESSIONAL SUMMARY

Senior Physical Design Engineer with 6 years of end-to-end RTL-to-GDSII experience across ultra-advanced FinFET and FD-SOI process nodes — 2nm (N2P), 3nm (N3P/N3A), 5nm, 7nm, GF22FDX, TSMC 12ffc, and GF 40nm. Proven track record of owning block-level and subsystem-level physical design closure for high-frequency, power-sensitive IPs delivered to Tier-1 customers including Google and ARM, with a documented achievement of 5GHz timing closure on glitch-detector critical paths. Expert in multi-corner multi-mode (MCMM) Static Timing Analysis (STA), power-grid integrity (static/dynamic IR drop), and Clock Tree Synthesis (CTS) optimization for low-skew, high-speed clock networks. Consistently accelerates tape-out schedules through Python/TCL-driven flow automation, reducing manual design-closure iterations and standardizing methodologies across large cross-functional teams.

KEY SKILLS

Communication | Teamwork | Problem Solving | Leadership | Adaptability | Critical Thinking | Decision Making

TECHNICAL SKILLS

Programming Languages: Python, TCL, Shell

Tools & Platforms: Fusion Compiler, Innovus, DC, Genus, ICV, LEC, PrimeTime, Tempus, RedHawk, Voltus, Calibre, Timing Closure, STA, DSO.ai,

CTS, Floorplanning, Physical Verification, DRC, Synthesis

Soft Skills: Communication, Teamwork, Problem Solving, Leadership, Adaptability, Critical Thinking, Decision Making, Time Management

Languages: English, Hindi, Bengali

WORK EXPERIENCE

Senior R&D Physical Design Engineer | Synopsys Inc. | Bangalore, IN (On-Site)

Sept 2024 – Present

- **Glitch Detector IP**
 - Led full RTL-to-GDSII physical design and PNR signoff and Delivered 5 GHz timing closure on critical N3P designs with custom clock tree strategies, achieving zero setup/hold violations and a fully signoff-clean GDSII for ARM. Drove TNS to zero via targeted ECO optimizations while meeting power/area budgets, enabling on-time tape-out to Google. Enhanced power integrity and closure efficiency through optimized grid design and automation, reducing IR drop (<5%) and improving TAT by 30% across N3P programs. Now working on N2P tech node with more architectural changes in the same IP. Fix LEC related issue and complex LUP violation, DRC/LVS and PERC clean.
- **SLM Subsystem Soft IP**
 - Owned full ownership of PVTC SIP releases which involves TOP/SDA/wrapper level STA signoff for PVTC Soft IP by authoring and maintaining SDC constraints, managing Design Compiler synthesis flows, and resolving complex customer JIRAs, ensuring clean netlist handoff to P&R with zero constraint regressions across five consecutive releases.
- **Process Detector IP**
 - Led full RTL-to-GDSII physical design and PNR signoff for high-speed SLM subsystems in the SF2A, N3A, N5 node, delivering a fully optimized, DRC/LVS-clean implementation with zero waived violations directly to Google, achieving all contractual PPA targets. Resolved 100% of routing DRC violations in a congestion-critical SLM subsystem by implementing an advanced cell spreading and layer assignment strategy during detailed routing, achieving clean Caliber signoff with zero remaining errors on tape-out. Built and deployed custom automation scripts for RO chain routing standardization, reducing per-block design closure time and enabling consistent power-grid implementation across N3P subsystem deliverables.

Engineer II ASIC Physical Design | AlphawaveSemi | Bangalore, IN (On-Site)

Jan 2024 – Sept 2024

- **Worked on Company's 1st Chiplet Implementation (company's 1st Chiplet product)**
 - Worked as Sub-System Level owner total RTL2GDS flow and interacting with Leaf-level block owners with leaf-level blocks' pin placement, dimensions calculation and guiding them on implementation
 - Fixed low node placement congestion by using different floorplan methods and manual placement, fixed leaf level block pin routing congestion in sub-system level
 - Worked closely with RTL and DFT team for constraint development
 - Independently worked on LEC issues, DRC, LVS cleaning, LDL & PERC
 - Help in leaf level blocks' timing fixing ECO, and Subsystem level timing signoff.

- **Responsible for GF - 22FDX Hierarchical block implementation, IO placement, Bump assignment and Timing closure**
 - Worked on various techniques of logical and physical synthesis to get best results on timing and area optimization. Analysing check_timing reports for constraint cleanup and LEC debugging and giving feedback to customer on RTL datatype.
 - Propose SDC changes to RTL owner based on analysis of complex clocking, Clock Domain Crossings and Reset Domain Crossings
 - Development of hierarchical block Level SDC by integrating block level SDC and aligning with Top Level Chip Architecture
 - Helped on Partitioning and Pin placement of leaf-level blocks while Floorplanning, while implementing 3.8 Million std cell instances with 500MHz clock frequency.
 - Worked on soft IP implementation and Hard IP implementation with automated custom powerplan for eMRAM, PLL, EFUSE, PVT Sensor as per integration guidelines
 - Implementation of custom clock tree with high drive strength buffers and flexible H-tree, using superbuffer implementation due to big block size in subsystem and Top level
 - Working with Customer and Vendor to Take Care of Integration Guidelines and constraints update provided by IP Vendor. IPs include eMRAM, PLL.
 - Responsible for ECO and Timing signoff, DRC, LVS Cleaning for final GDS release to TOP level.
 - Worked on IR violation fixed, and huge EM violation fixes in various techniques
 - Worked on Top level RDL routing and BUMP implementation with IO placement.

Engineer ASIC Physical Design Engineer | OpenFive (later known as AlphawaveSemi) | Bangalore, IN (On-Site)
Jan 2021 – Dec 2022

1. **Responsible for PnR and timing convergence of Memory Wrapper block in TSMC 12ffc**
 - Implementation from RTL to GDS.
 - Implementing memory Dense block and memory placement in 4mmX4mm block with 800MHz clock and custom clock tree build.
 - Timing closure by implementing few fixes to reduce cross talk.
 - Worked on Physical verification and IR analysis.
2. **Responsible for PnR and timing convergence of Memory dense block in GF 40nm**
 - Implementation from RTL to GDS.
 - Implementing standard cell Dense block and memory placement.
 - Worked on timing ECO manual fixed and IR analysis, Physical verification i.e., DRC/LVS/ANT.

Trainee Physical Design Engineer | Open-Silicon Research Pvt. Ltd (later known AlphawaveSemi) | Bangalore, IN (On-Site)
Jul 2020 – Dec 2020

- Worked in company's projects statistics with various projects data to generate a
- comparison for using better flavours of VT cell to get better PPA for different nodes
- 12nm, 7nm and 6nm.
- Responsible for Synthesis and PnR of PPA implementation
- Learning process of using commands and implementing Synthesis/PnR/PV from
- RTL2GDS.
- Worked on various tools to understand flow of PnR.
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EDUCATION & CERTIFICATIONS

EDUCATION:

BACHELOR OF TECHNOLOGY | Electronics & Communication Engineering
Jalpaiguri Government Engineering College (Autonomous)
DGPA: 7.44

2016– 2020

CERTIFICATIONS:

1. Physical Design Flow
2. CMOS Digital VLSI Design
3. Hardware Modelling using Python Core and Advanced Verilog
4. VT on Railway Communication System

PUBLICATIONS

A Comparative Study of Support Vector Machine and Naive Bayes Classifier for Sentiment Analysis on Amazon Product Reviews | Co-Author
Technologies / Tools Used: python, amazon database